FACULTY OF ENGINEERING AND TECHNOLOGY

BACHELOR OF TECHNOLOGY

DEEP LEARNING WITH N L P LABORATORY (303105480)

7th SEMESTER

COMPUTER SCIENCE & ENGINEERING DEPARTMENT

LABORATORY MANUAL

CERTIFICATE

This is to certify that Mr. **VAMSI VALLURI** with enrolment no. **2303031241366** has successfully completed his laboratory experiments in the **DEEP LEARNING WITH NLP LABORATORY (303105480)** from the department of **CSE-AI** during the academic year **2026-27**

Date of Submission: ....

Staff In charge: ....

Head Of Department: ....

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3.	Implementation to Convert the text to word frequency vectors with ScikitLearn (TfidfVectorizer).						
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PRACTICAL – 1

AIM: - Implementation of preprocessing of Text with NLTK (Tokenization, Stemming, Lemmatization and removal of stop words in NLP.

TOKENIZATION: -

Tokenization refers to break down the text into smaller units. It entails splitting paragraphs into sentences and sentences into words. It is one of the initial steps of any NLP preprocessing.

CODE: -

```
text = "Dog is walking on the road "

tokens = text.split()

print(tokens)
```

OUTPUT: -

```
#TOKENIZATION:
import nltk
```

```
text = "Dog is walking on the road "

tokens = text.split()

print(tokens)
```

```
['Dog', 'is', 'walking', 'on', 'the', 'road']
```

STEMMING: -

Stemming generates the base word from the word by removing the affixes of the word. It must be noted that stemmers might not always result in semantically meaningful base words.

CODE: -

```
import nltk
from nltk.stem import PorterStemmer
ps = PorterStemmer()
sentence = "Asif is cooking"
for word in sentence.split():
  print(ps.stem(word))
```

OUTPUT: -

```
import nltk
```

```
#STEMMING: -
from nltk.stem import PorterStemmer
ps=PorterStemmer()
sentence ="Asif is cooking"
for word in sentence.split():
  print(ps.stem(word))
```

```
asif
is
cook
```

LEMMATIZATION: -

Lemmatization involves grouping together the inflected forms of the same word. This way, we can reach out to the base form of any word which will be meaningful in nature. The base from here is called the Lemma.

CODE: -

import nltk

nltk.download('wordnet')

from nltk.stem import WordNetLemmatizer

lemmatizer = WordNetLemmatizer()

print(lemmatizer.lemmatize("cooking", pos= "v"))

print(lemmatizer.lemmatize("reading", pos= "v"))

OUTPUT: -

```
#LEMMATIZATION:
import nltk
#import spacy
nltk.download('wordnet')
from nltk.stem import WordNetLemmatizer
lemmatizer=WordNetLemmatizer()
print(lemmatizer.lemmatize("cooking", pos='n'))
print(lemmatizer.lemmatize("reading", pos='v'))
```

```
[nltk_data] Downloading package wordnet to /root/nltk_data...
cooking
read
```

REMOVAL OF STOP WORDS: -

Stop word removal is one of the most commonly used preprocessing steps across different NLP applications. The idea is simply removing the words that occur commonly across all the documents in the corpus.

CODE: -

```
import nltk

from nltk.corpus import stopwords

nltk.download('stopwords')

print(stopwords.words('english'))
```

OUTPUT: -

```
#Remove stop words
import nltk

from nltk.corpus import stopwords

nltk.download('stopwords')

print(stopwords.words('english'))

['a', 'about', 'above', 'after', 'again', 'against', 'ain', 'all', 'am', 'an', 'and', 'any', 'are', 'aren', "aren't", 'as', 'at', 'be', 'because', 'been', 'before', 'being',
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```